

REVIEW

Glia as architects of central nervous system formation and function

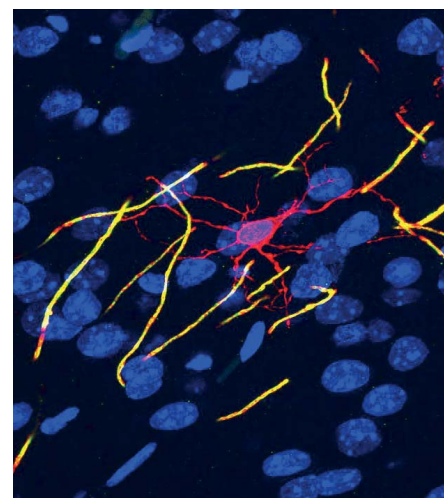
Nicola J. Allen^{1*} and David A. Lyons^{2*}

Glia constitute roughly half of the cells of the central nervous system (CNS) but were long-considered to be static bystanders to its formation and function. Here we provide an overview of how the diverse and dynamic functions of glial cells orchestrate essentially all aspects of nervous system formation and function. Radial glia, astrocytes, oligodendrocyte progenitor cells, oligodendrocytes, and microglia each influence nervous system development, from neuronal birth, migration, axon specification, and growth through circuit assembly and synaptogenesis. As neural circuits mature, distinct glia fulfill key roles in synaptic communication, plasticity, homeostasis, and network-level activity through dynamic monitoring and alteration of CNS structure and function. Continued elucidation of glial cell biology, and the dynamic interactions of neurons and glia, will enrich our understanding of nervous system formation, health, and function.

Cursed perhaps by the incorrect assertion of Virchow in 1846 that the brain contains a connective structure called “nervenkitt,” the mundane notion of glia as glue (derived from Greek) emerged and stuck [(1) and references therein]. This is despite the fact that soon after Virchow’s “discovery” of glia, the famous neuroanatomists of the late 19th and early 20th centuries had not only identified the major glia of the central nervous system (CNS) but also speculated with keen foresight on their potential diverse functions (1). Here we provide an overview of insights that are now addressing the question posed by Ramon y Cajal in 1909, “What is the function of glia?” [(1) and references therein]. We focus on the function of glia of the vertebrate CNS: radial glia, astrocytes, oligodendrocyte progenitor cells (OPCs, also called NG2 cells), oligodendrocytes, and microglia (Fig. 1). For example, we now know that radial glia are CNS progenitors that generate the majority of our neurons and glia, either directly or via inter-

mediate progenitors (Fig. 1). Astrocytes, star-shaped cells with thousands of processes that interact with likely all cell types of the CNS, exhibit a range of functions that help drive nervous system development and sculpt its activity (Fig. 2). OPCs are the most proliferative cells in the CNS, generate myelinating oligodendrocytes throughout life, and likely serve additional yet-to-be-identified roles in circuit formation and function (Fig. 3). Mature oligodendrocytes generate myelin sheaths that speed up nerve impulse conduction and provide metabolic support to axons (Fig. 3), and it now appears that dynamic regulation of myelination may regulate the precise timing of information propagation and communication across functional circuits. Finally, microglia are the guests of the CNS, best known as the brain’s resident macrophages, but with increasingly evident roles during multiple stages of nervous system development and activity (Fig. 4).

One of the most quoted “facts” about glia is that they outnumber neurons by 10 to 1 in the human brain, but it is now clear that we have a roughly equal proportion of neurons and glia [(2) and references therein]. The relative proportions vary by region (e.g., gray matter versus white matter), developmental stage, and species, but as a very general rule, human brains are composed of 20% astrocytes (2), 3 to 10% OPCs (3), 25% oligodendrocytes (2),



Myelinating oligodendrocyte in the mouse cortex

and 5 to 15% microglia (4). We have known of neuronal diversity for more than 100 years, and it is now emerging that glia also exhibit important functional diversity [(5–7) and references therein], likely to be driven by both intrinsic programs based on developmental origins and extrinsic interactions with their environment. In this review, we take a chronological perspective that aims to chart how glia first drive neurogenesis and subsequently regulate neuronal migration, axon growth, synapse formation, and, ultimately, circuit function. We focus on the role of glia and their reciprocal interactions with neurons in regulating healthy nervous system formation and function and refer interested readers to other excellent sources on the development of glia (3, 8–10) and the myriad roles of glia in disease (4, 11–13).

Glia in CNS formation

Glia are the principle regulators of cell number in the CNS

During embryonic development, neuroepithelial progenitors take on characteristics of astrocytes, including the expression of factors such as glutamate transporters, glycogen granules, and intermediate filaments. These neuroepithelial cell-derived “radial glia” go on to generate neurons during embryonic stages of mammalian development [(14) and references therein] and then switch to generating

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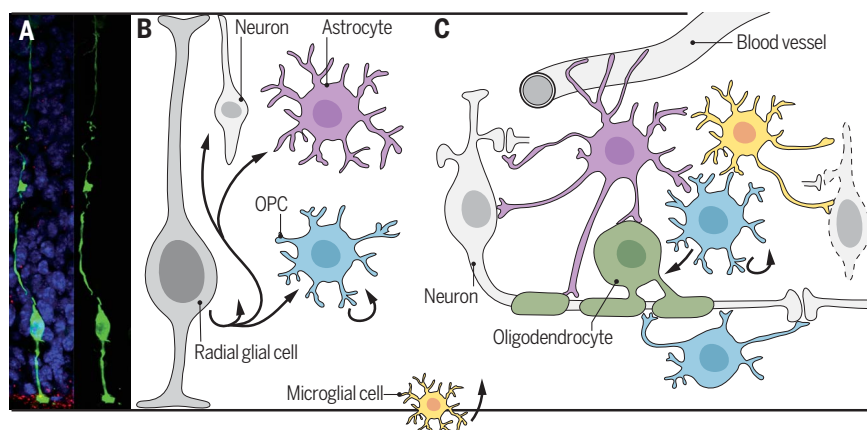


Fig. 1. Origin and overview of CNS glial cells.

(A) Confocal image of a radial glial cell in an E14 mouse cortex, visualized after in utero electroporation of green fluorescent protein (GFP, green) and costaining for apical centrosomes (pericentrin, red) and cell nuclei [4',6-diamidino-2-phenylindole (DAPI), blue] (left). [Photo credits: Sven Falk and Magdalena Goetz, Helmholtz Centre, Munich] (B) Radial glial cells are the principal neuroepithelial progenitor cells of the central nervous system and generate the majority of CNS neurons and glia, either directly (e.g., neurons) or indirectly through intermediate progenitors (e.g., OPCs). Microglia (yellow) enter the CNS during embryonic development. (C) Neurons and glia interact in a myriad of ways (explained in text and subsequent figures). Dashed line indicates dying neuron.

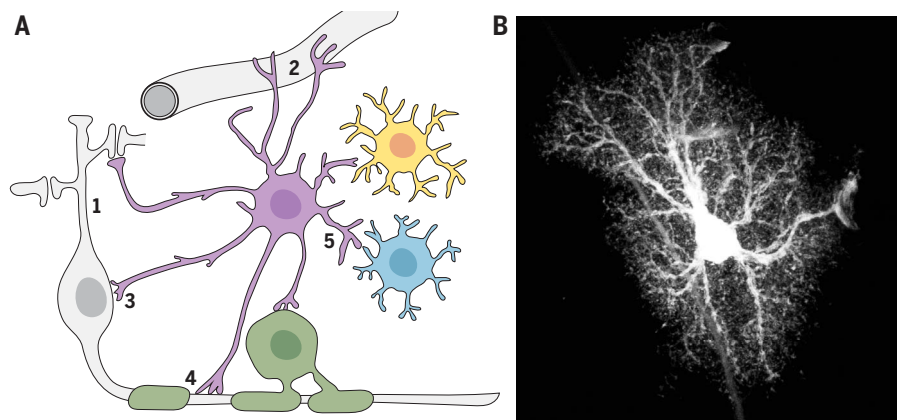


Fig. 2. Astrocytes. (A) Astrocytes (purple) have a characteristic star-like morphology and send out multiple branches that terminate in thousands of fine processes that interact with synapses, blood vessels, and other cells. Astrocytes regulate synapse formation, elimination, and function (1); have endfeet that ensheath CNS vasculature (2); contribute to metabolic support and homeostatic function; and have reciprocal interactions with neurons regulating circuit function (3). Astrocytes also make contact with nodes of Ranvier (4), the function of which is unclear, and interact bidirectionally with OPCs, oligodendrocytes, and microglia (5), the relevance of which is emerging. (B) Astrocyte confocal image. [Photo credit: The Cell Image Library, image CIL 48001; available at www.cellimagelibrary.org/images/48001]

mature glia, which they do either directly by generating proliferating astrocytes or through generation of intermediate progenitors, such as OPCs [(3) and references therein]. Toward the end of gliogenesis, many radial glia differentiate directly into mature astrocytes, whereas in anamniotic vertebrates (e.g., fish and amphibians), mature radial glia retain a radial morphology but may subserve the functional roles of astrocytes (15). Subtypes of radial glia with distinct neurogenic and gliogenic potentials have been described, and radial glia-derived progenitors persist through adulthood and give rise to adult neural stem cells [(16) and references therein]. In addition to cells derived from the neuroepithelium, microglia, which are derived from erythromyeloid progenitors, enter the vertebrate CNS early in development at the onset of neurogenesis and subsequently proliferate and migrate to colonize the entire brain and spinal cord [(10) and references therein]. Microglia regulate neuronal survival, phagocytose excess neurons undergoing apoptosis during early development and in neurogenic regions of the adult brain [(17) and references therein], and have multiple roles in refining CNS formation and function [(18) and references therein]. Although glia regulate neuron number (14), neurons can, in turn, regulate glial cell number. For example, the programmed cell death of specific neurons influences initial microglial cell entry to the CNS [(18) and references therein], and neuronal activity can regulate OPC proliferation and oligodendrocyte generation [(19) and references therein]. As will become clear, many aspects of nervous system formation and function are underpinned by dynamic bidirectional interactions of neurons and glia.

Glia influence neuronal migration, axon specification, and growth

Following their birth, neurons in the CNS migrate, either radially through the neuroepithelium—for example, establishing specific cortical layers—or

tangentially, as exemplified by interneuron migration from the ganglionic eminences to the cortex. The role of glia in regulating tangential migration has not been intensively studied, although microglia have been implicated [(10) and references therein]. By contrast, it is well known that radial glia support radial migration of neurons, and the underlying mechanisms have been extensively investigated [(14) and references therein]. For example, disruption of radial glial cell polarity dysregulates neuronal migration to such an extent that a duplicated cortex can form underneath a normal one (14). Radial glia also exhibit polarized mRNA transport and localized protein translation along their processes (20), suggesting that they may locally regulate additional aspects of neuronal development. Indeed, axon specification occurs on the opposite side of the neuron to that of radial glial contact, at least in vitro (21). In addition, microglia (10), astrocytes, and OPCs (22) have been shown to subsequently help guide axons to their destinations, and analogous roles for glia in axon specification (23) and guidance have also been identified in *Caenorhabditis elegans* (24).

Glia can coordinate circuit-wide neuronal differentiation

Our understanding of the mechanisms that coordinate neuronal differentiation across circuits is rudimentary, but it is now clear that glial cells can coordinate the timing of axon-driven target neuron differentiation. In the *Drosophila* visual system, photoreceptor neurons first secrete a signal to their associated wrapping glia (analogous to myelinating glia), which then secrete cues that induce target neuron differentiation. This indirect signaling via glia introduces a delay that controls the timely differentiation of target neurons, because the wrapping glia arrive at specific target locations with a delay relative to axons (25). The mechanisms that might coordinate differentiation of neurons distributed across local

and long-range circuits in vertebrates are unclear. However, given that invertebrate models have served as mechanistic pathfinders for many aspects of vertebrate neural development [(26, 27) and references therein], the prospective role of glia in regulating circuit-level neuron differentiation warrants further investigation.

Glia regulate synapse formation and pruning

After neurogenesis, neuronal migration, and axon guidance are complete, prospective partner neurons begin to make synaptic connections. Synaptogenesis involves a series of formation, strengthening, and remodeling steps, with glia playing distinct roles at each stage. The timing of initial synaptogenesis coincides with the generation of astrocytes, putting them in the right place at the right time [(28) and references therein]. First, astrocytes contact immature neurons, making them competent to form synapses (28). Then, neurons become receptive to astrocyte-secreted signals that instruct formation of immature synapses, including thrombospondins, which induce silent synapses, and glypicans, which induce functionally active synapses (28). Synapses are then matured by additional glial signals, including Sparc1 from astrocytes, which induces synaptic strengthening and stabilization (18). As a balance to the prosynaptogenic signals, astrocytes also produce signals (e.g., Sparc) that negatively regulate synapse formation and function [(18) and references therein]. In addition to astrocytes, microglia have been implicated in synapse formation [(18) and references therein]. The various roles of distinct glia in regulating synapse formation and the extent to which they do so throughout the nervous system and over time in vivo remains to be fully determined.

Toward the end of the synaptogenic period, weak and inappropriate synapses are eliminated by astrocytes and microglia, ultimately leaving neurons with their adult connectivity. Astrocytes phagocytose synapses via Megf10- and Mertk-driven phagocytic pathways, whereas microglia prune synapses tagged with components of the classical complement pathway [(18) and references therein] or through the Trem2 pathway (29). Interleukin-33 signaling from astrocytes to microglia has also been shown to regulate synapse elimination (30), highlighting the complexity of cell-cell interactions in the CNS.

A role for glia in regulating synapse development appears broadly conserved across evolution. For example, human astrocytes transplanted into mice can regulate multiple aspects of the structure and function of host synapses [(5) and references therein]. Invertebrate glia also regulate synapses, with glia in *C. elegans* determining the precise location of specific synapses and promoting synaptogenesis (27) and *Drosophila* glia regulating synapse maturation (26). The distinct roles of glia in synapse formation and remodeling is likely to be regulated in part by their regional identity (8) and complex interactions with their environment. For example, neurons can regulate astrocyte maturation and diversity

in vivo (28), and neural activity can regulate hundreds of astrocyte genes in vitro (31). Therefore, distinct neurons may be able to influence and use specialized glial functions to sculpt circuit-specific architecture. Determining how interactions between neurons and glia regulate different synaptic classes in different brain regions in vivo will be an important area of exploration.

Glial in regulating CNS function

Once neuronal circuits are assembled, glia regulate numerous aspects of nervous system function. They do so by affecting synaptic function, ion homeostasis, and metabolism and likely also through yet-to-be-identified roles.

Glial adjust synaptic communication and plasticity

Glial regulate many aspects of synaptic function. Astrocytes regulate neurotransmitter uptake, for example, via glutamate transporters [(32) and references therein], which is dynamically regulated; glutamate transporters on astrocyte processes are up-regulated by high activity and down-regulated during prolonged periods of low activity, providing a way to modulate the concentration of transmitter available at synapses (32). Furthermore, increased synaptic activity can also lead to a greater ensheathment of synapses by astrocyte processes, decreasing the chance of neurotransmitters diffusing and activating a neighboring synapse [(33) and references therein]. Astrocytes express receptors for multiple neurotransmitters and neuromodulators. Many of these are G protein-coupled receptors associated with intracellular signaling cascades that converge on intracellular calcium ion (Ca^{2+}) signaling [(33) and references therein]. Astrocytes can, in turn, release gliotransmitters, thus signaling back to

neurons and other glial cells [(34) and references therein]. Whether gliotransmitter release primarily regulates the function of individual synapses or groups of synapses (given that an individual astrocyte can contact thousands of synapses) is likely to depend on the synapse type and gliotransmitter in question. Gliotransmitters include adenosine triphosphate, which, when converted to adenosine, acts on presynaptic adenosine receptors to alter neurotransmitter release probability and short-term plasticity. In addition, D-serine, released from astrocytes in a calcium-dependent manner, is a coagonist for the N-methyl-D-aspartate (NMDA) glutamate receptor, through which it can also regulate synaptic plasticity [(32–34) and references therein]. Astrocyte activation has even been shown to enhance both neuronal plasticity and memory in vivo (35).

In addition to astrocytes, OPCs and microglia have also been shown to regulate synaptic function. For example, OPCs have been implicated in controlling postsynaptic NMDA and AMPA receptor-mediated currents and plasticity through cleavage of the transmembrane proteoglycan NG2, and NG2 mutant mice exhibit deficits in sensorimotor function [(19) and references therein]. Dysregulation of microglial signaling also affects synaptic and circuit function. For example, cell type-specific knock out of purinergic receptors leads to disruption of ocular dominance plasticity in the visual cortex (4), and tumor necrosis factor- α , a cytokine secreted by microglia that regulates homeostatic synaptic scaling, influences ocular dominance plasticity (36). In addition, depletion of microglia from the juvenile mouse brain inhibits motor learning, with microglial brain-derived neurotrophic factor proposed to mediate the effect. However, the elimination of microglia at adult ages does not affect behavior (4, 18), further indi-

cating that stage- and region-specific neuron-glial interactions regulate neural circuit function.

Glial regulation of ion homeostasis affects circuit function

Glial can affect neuronal excitability by regulating ion homeostasis, for example, by clearing potassium (K^+) ions from the extracellular space, which facilitates neuronal membrane repolarization after an action potential and thus continued firing (37). The glial cells responsible for K^+ uptake can vary by brain region, and these include astrocytes, OPCs, and oligodendrocytes [(37) and references therein]. Along myelinated axons, most K^+ channels are localized under the myelin sheath, suggesting that K^+ clearance might occur at the interface of the axon and the myelin sheath, rather than at the node of Ranvier where astrocytes and OPCs make contact. Indeed, knock out of the inward-rectifying potassium channel Kir4.1 from mature myelinating oligodendrocytes leads to poor K^+ clearance and consequent conduction defects (38), revealing a function for mature oligodendrocytes in facilitating circuit function.

Glial regulation of extracellular K^+ can influence diverse aspects of neural circuit function. For example, in the rat lateral habenula (LHb), Kir4.1 is expressed at low levels in astrocytes but is up-regulated in models of depression. Regulation of astrocyte Kir4.1 can alter both the bursting activity of LHb neurons and associated depression-like symptoms (39). By contrast, in the spinal cord, the astrocytes surrounding large, fast motor neurons express a very high level of Kir4.1, which is necessary to maintain the high firing rate of these specific neurons (40). In the cortex, specialized satellite oligodendrocytes tightly associate with the cell bodies of cortical neurons and regulate extracellular K^+ concentration and

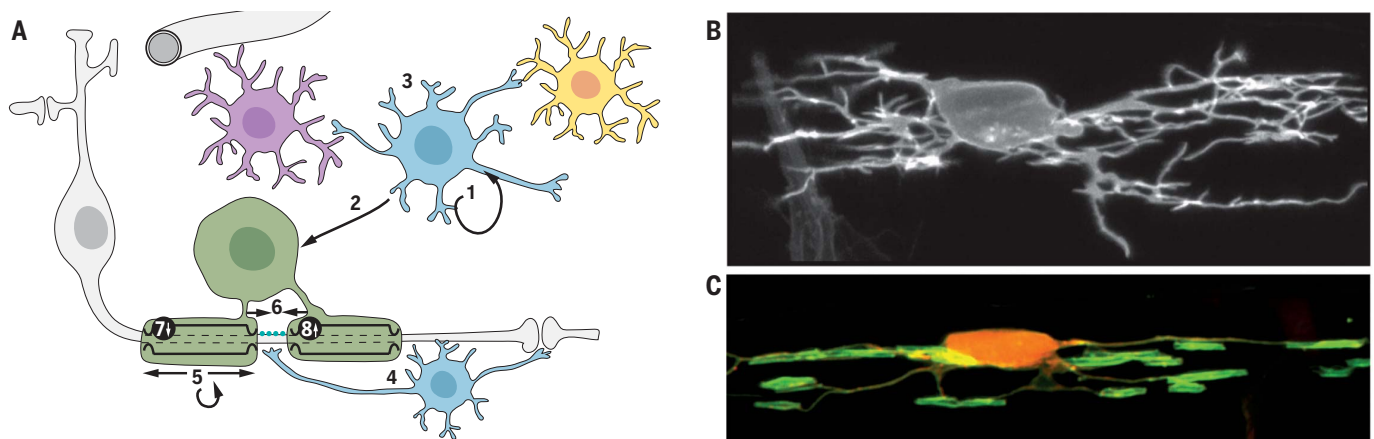


Fig. 3. OPCs and oligodendrocytes. (A) OPCs (blue) are the most proliferative cells of the CNS (1) and generate mature myelinating oligodendrocytes throughout life (2). OPCs interact with many other cells of the CNS (3), particularly in disease. OPCs extend processes that contact nodes of Ranvier, receive synapses from axons, and regulate synaptic function (4). Whether there are distinct subtypes of OPCs or simply different functional states remains unclear. Oligodendrocytes (dark green) produce lipid rich myelin sheaths that wrap around axons and regulate action potential conduction velocity. Dynamic regulation of myelination in response to neuronal signals (5) may be a fundamental mechanism by which neural circuit function is fine-

tuned. Myelinating oligodendrocytes organize axonal domains (6), including nodes of Ranvier, where the voltage-gated Na^{2+} channels that mediate action potential propagation are localized; provide metabolic support to axons (7); and facilitate ion homeostasis that is essential to normal action potential conduction, for example, the uptake of K^+ ions by myelin (8). (B) Confocal image of an OPC expressing membrane-tethered fluorescent protein visualized in a living transgenic zebrafish larvae at 3 days postfertilization. (C) Confocal image of an oligodendrocyte expressing membrane-tethered GFP (green) and cytoplasmic red fluorescent protein (orange) visualized in a living transgenic zebrafish larvae at 4 days postfertilization. [Photo credit: Marion Baraban, Lyons lab]

neuronal activity through formation of a glial syncytium with neighboring astrocytes and oligodendrocytes [(19) and references therein]. It remains unknown to what extent region-specific glial phenotypes (e.g., high or low Kir4.1 expression) are driven by intrinsically distinct developmental programs or by dynamic interactions with the extrinsic environment.

Neuro-glial-vascular coupling provides metabolic support

Interactions between the vasculature and cells of the CNS are increasingly implicated in brain health. CNS blood vessels have specialized barrier properties, known as the blood-brain barrier, which limit movement of substances, including nutrients, between the blood and the brain (41). However, astrocytes are ideally positioned to actively supply neurons with nutrients because they extend processes that contact both blood vessels and neurons (42). Astrocytes are not just relays but also process precursor molecules for delivery to neurons. For example, astrocytes are enriched in lipid synthesis pathways and provide lipids to neurons, including cholesterol, which is essential for maintaining pre-synaptic function (28). Astrocytes are also enriched in glycolytic pathways and make glycogen, which can be converted into lactate and transferred to neighboring neurons by monocarboxylate transporters (43). Although there has been debate about the specific role of lactate as an energy source for neurons, mice lacking monocarboxylate transporters do show impairments in long-term memory formation (43).

Along myelinated axons, myelin sheaths create a barrier between the axon and the extracellular environment, limiting direct metabolic support routes to the axon to the short unmyelinated gaps between myelin sheaths at nodes of Ranvier. Instead, myelinating oligodendrocytes generate glycolytic products, including lactate, which can be transported to the axon via monocarboxylate transporters at the myelin-axon interface (44).

It has been proposed that high-frequency axonal firing activates NMDA receptors on oligodendrocytes, triggering increased glucose uptake and the subsequent transport of lactate to the axon to support the energy requirements associated with maintaining high levels of activity (44). Visualizing this metabolic support system in real time and in response to neural activity and investigating how astrocytes and oligodendrocytes collaborate on metabolic support are important areas of ongoing investigation.

Glia regulate network-level function Astrocytes dynamically alter brain states

There is now compelling evidence of glial involvement in dynamic regulation of circuit-level function. For example, networks of astrocytes in the mouse visual cortex can be activated by the neuromodulator norepinephrine in line with dynamic changes in the arousal state of the animal. This increased astrocyte activity can in turn enhance the ability of astrocytes to detect altered neuronal activity (5). The regulation of circuit function via astrocytes has also been demonstrated in *Drosophila*, where octopaminergic neurons signal through astrocytes to control downstream neurons and behavior (28). Furthermore, dynamic regulation of astrocyte population-level Ca^{2+} activity has been shown to alter ensemble-level oscillatory states in mice (5). More than a century ago, Ramon y Cajal proposed that astrocytes may regulate sleep state, a suggestion now supported by extensive evidence (5).

Myelination dynamically alters circuit function

Myelination of axons by oligodendrocytes has long been known to increase conduction velocity. In addition to whether an axon is myelinated or not, changes to the number, distribution, length, and thickness of myelin sheaths all, in principle, affect conduction velocity and thus the timing of information propagation, prompting speculation

that dynamic regulation of myelination in response to neural activity may fine-tune the timing of communication between neurons in a circuit. Indeed, neural activity can regulate many aspects of oligodendrocyte lineage progression in developing, juvenile, and adult animals [(19) and references therein]. OPCs receive bona fide synapses from neuronal axons, which may mediate the effects of neural activity on their proliferation and/or differentiation (19). Mature myelinating oligodendrocytes also express a range of neurotransmitter receptors, ion channels, and transporters, whose diverse functions remain to be fully elucidated but which likely mediate the effects of neural activity on myelination itself. That only some neuronal subtypes regulate myelination via activity [(19) and references therein] suggests that activity-regulated myelination may be circuit specific.

Direct evidence that activity-regulated myelination, in turn, affects circuit function comes from several studies. Social isolation of mice, for a distinct postnatal period or a prolonged period in adulthood, disrupts myelination in the prefrontal cortex [(19) and references therein]: In addition, the behavioral traits associated with social isolation can be phenocopied by loss of oligodendrocyte ErbB receptor signaling during the corresponding postnatal period and reversed by chemically promoting myelination in adulthood [(19) and references therein]. These results indicate that the changes in neural activity associated with isolation lead to behavioral deficits via changes in myelination. In addition, learning new motor tasks promotes new oligodendrocyte generation in adult mice, and the prevention of such oligodendrogenesis impairs learning [(19) and references therein]. Nonetheless, it remains to be determined how dynamic regulation of myelination affects the timing of conduction, synaptic communication, and the function of neuronal populations within intact circuits in vivo (19). Evidence that neurons regulate myelinating glia, and that myelinating glia, in turn, regulate circuit function, further exemplifies how reciprocal interactions between neurons and glia play fundamental roles in circuit function.

Conclusions

Here we have highlighted integral roles of glial cells in many aspects of CNS formation and function, and we refer interested readers to our more extensive supplementary reading list for further information. Given the emerging evidence of diverse subtypes and dynamic states of glia, we expect that further new functions will be revealed, and their mechanistic underpinnings elucidated, in years to come. This will require interrogation using a variety of models and approaches, bridging of scales from molecule to system, and investigating mechanisms from development through the entire life course. The study of glia reminds us that nervous system development does not end at birth, nor indeed at adolescence, and that the dynamic behavior of glial cells and ongoing neuron-glial interactions actively sculpt and remodel the CNS throughout life. We expect that further insight into how the brain forms, functions, and ages in a healthy

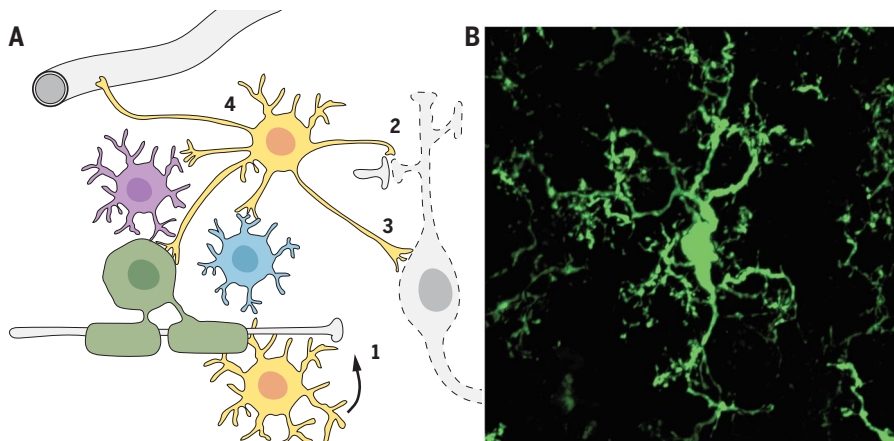


Fig. 4. Microglia. (A) Microglia (yellow) are the resident immune cells of the brain, entering during early development from the periphery (1). In addition to immune surveillance roles (not shown), microglia interact with multiple cell types of the CNS and regulate numerous developmental and functional processes, including synaptic pruning (2), clearing apoptotic neurons (3), and interacting with multiple CNS cell types, in health and disease (4). (B) Confocal image of microglia expressing GFP in mouse cortex (Cx3cr1-GFP). [Photo credit: Youtong Huang and Greg Lemke, Salk Institute]

manner will derive from integrating our understanding of how neurons interact with glia, how different glial cells interact with each other, and how the brain interacts with the rest of the body.

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